

Toni J.B. Liu

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EDUCATION

Cornell University, Ithaca, NY

August 2022 – Present

Ph.D. Candidate in Physics | GPA: 4.09 / 4.3

Advisor: Christopher J. Earls

California Institute of Technology, Pasadena, CA

September 2019 – June 2021

Bachelor of Science: Applied Physics | GPA: 3.84 / 4.3

Wesleyan University, Middletown, CT

September 2016 – May 2019

Bachelor of Arts: Physics and Film Studies | GPA: 3.75 / 4.3

RESEARCH INTERESTS

Large language models, mechanistic interpretability, LLM scaling laws, manifold learning, information geometry

HONORS AND AWARDS

Mong Cornell Neurotech Fellowship, Cornell University

August 2023 – August 2024

Larson Scholarship, California Institute of Technology

Summer 2020

Dean's List, Wesleyan University

Fall 2016, Fall 2017, & Spring 2018

SELECTED PUBLICATIONS

- **T. J.B. Liu**, J. Bao, Y. Liu, G. Arora, N. Boullé, R. Sarfati, & C. J. Earls, The Geometry of Ignorance: LLMs Know When to Temper Bayesian Priors, under review at ICLR 2027
- **T. J.B. Liu**, B. Zadeoglu, N. Boullé, R. Sarfati, & C. J. Earls, Jacobian Scopes: token-level causal attributions in LLMs, under review at NeurIPS 2026 ([arXiv:2601.16407](https://arxiv.org/abs/2601.16407))
- **T. J.B. Liu**, N. Boullé, R. Sarfati, & C. J. Earls, Density estimation with LLMs: a geometric investigation of in-context learning trajectories, ICLR (2025)
- **T. J.B. Liu**, N. Boullé, R. Sarfati, & C. J. Earls, LLMs learn governing principles of dynamical systems, revealing an in-context neural scaling law, Oral presentation, EMNLP (2024)
- T. Yu* & **T. J.B. Liu***, A. Tseng & C. De Sa, Shadow cones: a generalized framework for partial order embedding, ICLR (2024) (Equal contribution)
- J. Bao, N. Boullé, **T. J.B. Liu**, R. Sarfati, & C. J. Earls, Text-trained LLMs can zero-shot extrapolate PDE dynamics, revealing a three-stage in-context learning mechanism, Machine Learning: Science and Technology (2026)
- R. Sarfati, **T. J.B. Liu**, N. Boullé, & C. J. Earls, Lines of Thought in Large Language Models, ICLR (2025)
- A. Tseng, T. Yu, **T. J.B. Liu**, & C. De Sa, Coneheads: hierarchy-aware attention, NeurIPS (2023)

RESEARCH EXPERIENCE

Theory of data scaling laws, Gore Laboratory, MIT

June 2026 – Present

- Extending Jacobian Scopes to quantify the internal compute induced by each training token, separating tokens into contemplative and no-brainer groups
- Investigating the scaling relation between token pensiveness and training repetition to improve the training efficiency of LLMs

Mechanistic interpretation of LLMs, Earls Group, Cornell University

May 2025 – Present

- Developed Jacobian Scopes, a suite of fast, first-order methods that attribute an LLM's prediction to its inputs
- Discovered the “direction of ignorance”, a geometric feature that encodes unigram statistics that LLMs default to when uncertain

Dynamics of in-context learning, Earls Group, Cornell University

August 2023 – Present

- Built Hierarchy-PDF, a computationally efficient framework to extract statistical information of dynamical systems learned by transformer-based LLMs
- Demonstrated LLMs' zero-shot ability to model the evolution of dynamical systems without fine-tuning or instruction prompting
- Discovered an in-context neural scaling law, relating the fidelity of learned transition rules to the number of states observed in-context
- Geometrically bounded the in-context learning trajectories using information geometry and optimal kernel theory

Embedding and attention in non-Euclidean spaces, Relax ML Lab, Cornell University *March 2023 – December 2024*

- Developed Shadow Cones, a fast framework for embedding concept graphs in hyperbolic space
- Derived Coneheads, a hierarchy-aware alternative to dot-product attention that empirically improved task performance and parameter efficiency across language, vision, and graph tasks

Memory–abstraction trade-off in diffusion models, Earls Group, Cornell University *February 2023 – August 2024*

- Developed diffusion RNN: a fully recurrent neural network that uses a reverse-diffusion process to extract low-dimensional structures in data as quasi-stable manifolds
- Investigated the memory-abstraction trade-offs in Diffusion RNN

Energy-based anomaly detection, Cohen Lab, Cornell University *January 2023 – July 2023*

- Developed energy-based machine learning algorithms to automatically detect and correct anomalies in reconstructed flight trajectories of insects

Decoding latent time-keeping circuits in algae, Meyerowitz Lab, Caltech *June 2020 – January 2023*

- Introduced a variational-autoencoder approach to discover latent dynamical states from imaging data, revealing a low-dimensional time-keeping mechanism
- Interpreted the resulting latent-state dynamics with PDE and Kuramoto oscillator models, connecting self-organized waves to anticipatory responses under periodic stimuli

Optical characterization of phase-change materials, Sher Lab, Wesleyan University *February 2018 – July 2019*

- Built COMSOL thermo-optical simulations and ellipsometry models to characterize the temperature-dependent optical constants of GST and ZnO in photonic limiters

TEACHING

Phys 2207 – Newtonian Mechanics and Fluid Mechanics, TA and lab instructor, Cornell Univ. *Fall 2022*

Phys 2208 – Electricity, Magnetism and Quantum Mechanics, TA and lab instructor, Cornell Univ. *Spring 2023*

SKILLS

Machine Learning: PyTorch, Hugging Face, NumPy, SciPy, CUDA, Pandas

Programming Languages: Python, MATLAB, Mathematica, COMSOL Multiphysics, LabVIEW, Git

Visual Presentation: Adobe Photoshop, Final Cut Pro, Blender, Apple Motion, matplotlib

Languages: English (fluent), Mandarin (native), Italian (working proficiency)

PERFORMING AND MEDIA ARTS

Film-maker and Animator, Wesleyan Cardinal Pictures and Independent *February 2017 – Present*

- Directed short films OCD – A Love Story and Allegory of the Grotto, both screened at the Wesleyan Student Film Festival; currently developing the animation series Life: a Study of Motion

Tenor, Cornell University Chorale *February 2023 – Present*